NOTE The document identifier and heading has been changed on this page to reflect that this is a performance specification There are no other changes to this document The document identifier on subsequent pages has not been changed, but will be changed the next time this document is revised

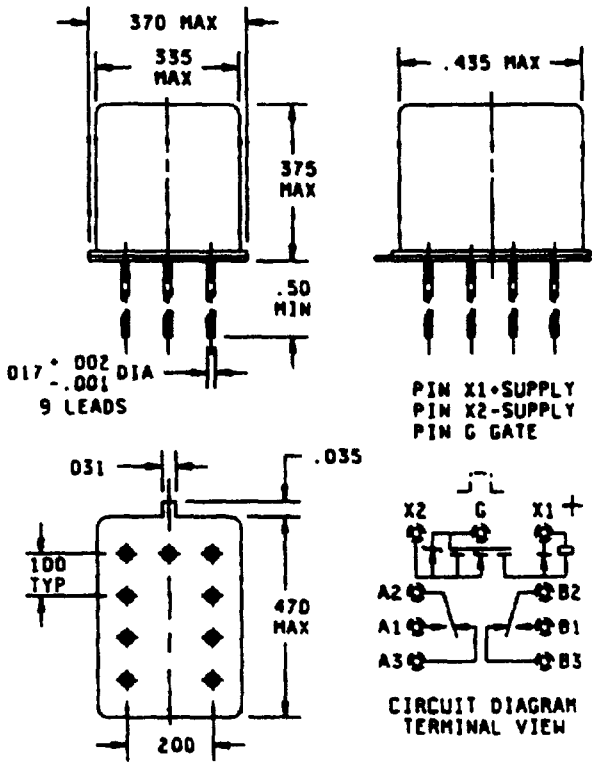
MIL PRF 28776/7D
26 January 1989
SUPERSEDING
MIL-R-28776/7C
4 May 1988

PERFORMANCE SPECIFICATION SHEET

RELAYS, HYBRID, ESTABLISHED RELIABILITY, DPDT
LOW LEVEL TO 1 0 AMPERE
(SENSITIVE COIL OPERATE POWER AT 25°C), INTERNAL MOSPET DRIVER
WITH ZENER DIODE GATE PROTECTION (ELECTROMECHANICAL OUTPUT),
DIODE COIL SUPPRESSION AND TERMINALS WITH 0 100 GRID LEAD SPACING

This specification is approved for use by all Departments and Agencies of the Department of Defense

The requirements for acquiring the product described herein shall consist of this specification sheet and the issue of the following specification listed in that issue of the Department of Defense Index of Specification and Standards (DODISS) specified in the solicitation MIL-R-28776



○ [illegible]

(D) denotes changes

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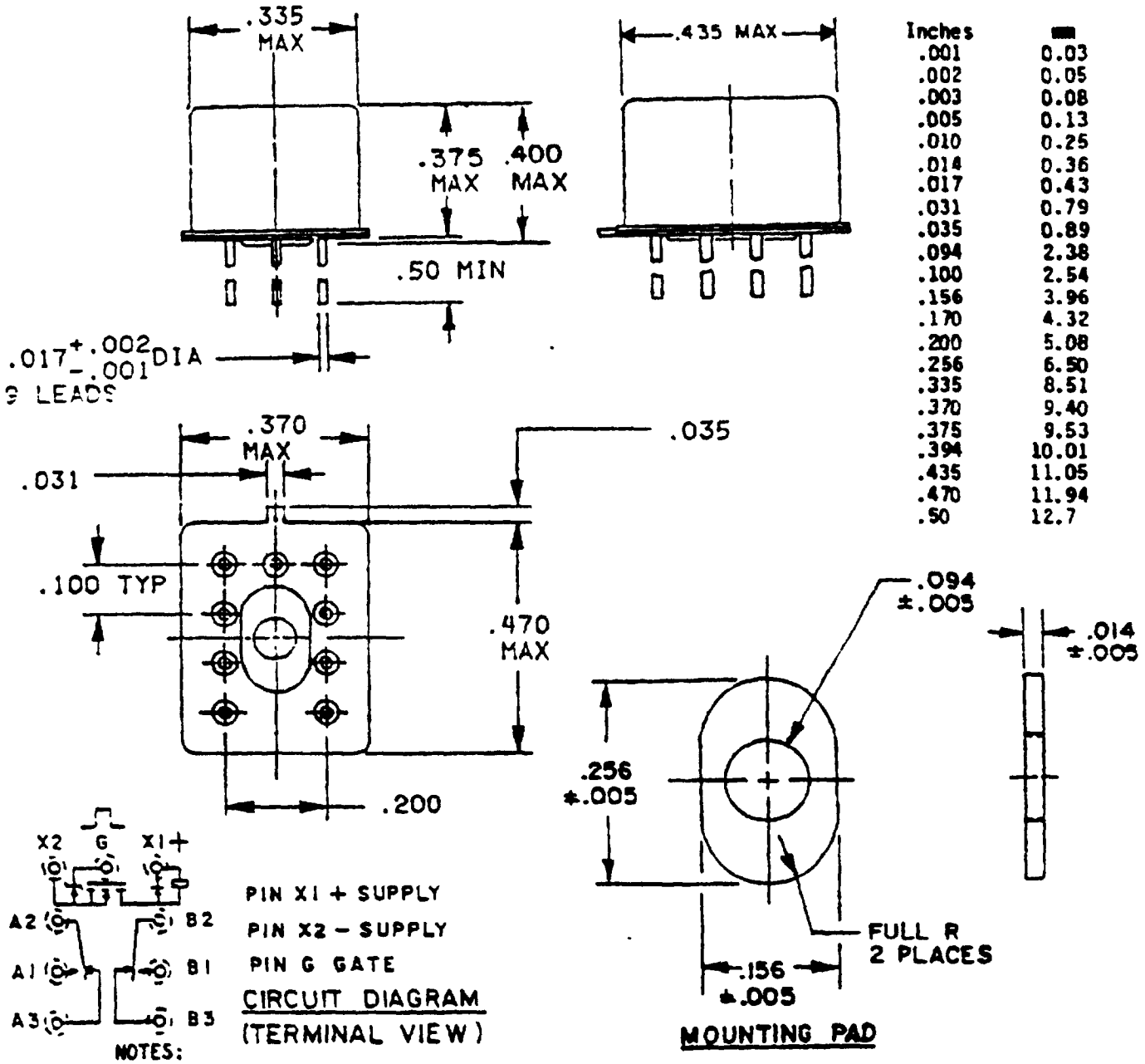
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Inches	mm
.001	0.03
.002	0.05
.003	0.08
.010	0.25
.017	0.43
.031	0.79
.035	0.89
.100	2.54
.200	5.08
.335	8.51
.370	9.40
.375	9.53
.435	11.05
.470	11.94
.500	12.70

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Unless otherwise specified, tolerance is ±.010 (0.25 mm).
4. Terminal numbers shown above are for reference only. Numbers do not appear on relay.
5. Relays shall have a plus (+) sign placed on the circuit diagram as shown.
6. Coil symbol optional in accordance with MIL-STD-1285.
7. Schematic diagram shown on part is the terminal view.

FIGURE 1. Dimensions and configuration.- Continued.

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Inches	mm
.001	0.03
.002	0.05
.003	0.08
.005	0.13
.010	0.25
.014	0.36
.017	0.43
.031	0.79
.035	0.89
.094	2.38
.100	2.54
.156	3.96
.170	4.32
.200	5.08
.256	6.50
.335	8.51
.370	9.40
.375	9.53
.394	10.01
.435	11.05
.470	11.94
.50	12.7

CIRCUIT DIAGRAM
(TERMINAL VIEW)

MOUNTING PAD

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Unless otherwise specified, tolerance is ±.010 (0.25 mm).
4. Terminal numbers shown above are for reference only. Numbers do not appear on relay.
5 Relays shall have a plus (+) sign placed on the circuit diagram as shown.
6 Coil symbol optional in accordance with MIL-STD-1285.
7. Circuit diagram shown on part is the terminal view.
8. Mounting pad shall be a polyester film in accordance with MIL-I-631, type G, class 1 Mounting pad shall be rigidly attached to the relay as shown.

(D) FIGURE 2. Dimensions and configuration (relay with mounting pad).

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REQUIREMENTS.

Input data:

Supply voltage: See table I.

Gate turn-off voltage (or current): 0.5 V dc maximum.

Gate turn-on voltage. 4.3 V dc minimum, 15 V dc maximum.

Zener voltage: 20 ±3 V dc over the temperature range.

Coil transient suppression: 1.5 V dc maximum.

Turn-off time. 7.5 milliseconds maximum.

Turn-on time 4.0 milliseconds maximum.

Voltage transients: Applicable.

Diode peak inverse voltage: 100 V dc minimum.

Contact data:

Load ratings:

High level (relay case grounded):

Resistive.

1.0 ampere at 28 V dc.

500 milliamperes at 115 V ac 400 hertz case not grounded.

250 milliamperes at 115 V ac 60 hertz case not grounded.

100 milliamperes at 115 V ac 60 and 400 hertz case grounded.

Inductive: 0.2 ampere at 28 V dc with 0.32 henry inductance.

Lamp: 0.10 ampere at 28 V dc.

Low level: 10 to 50 microamperes at 10 to 50 millivolts dc or peak ac maximum.

Intermediate current: Applicable.

Contact resistance or voltage drop:

Initial: 0.10 ohm maximum (tested at 10 milliamperes maximum at 6 V dc maximum or peak ac).

High level:

During life: Not more than 5 percent of open circuit voltage

After life: 0.20 ohm maximum.

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Intermediate current:

During intermediate current: 1.0 ohm maximum.

After intermediate current: 0.20 ohm maximum.

Low level:

During life: 100 ohms maximum.

After life: 0.15 ohm maximum.

Contact bounce: 1.5 milliseconds maximum (applicable to failure rate level "L").

Contact stabilization time: 2.0 milliseconds maximum (applicable to failure rate levels "M", "P", and "R").

Overload (high level only): Two times rated current.

Electrical data:

Insulation resistance: 1/ 10,000 megohms at 500 V dc except the resistance between coil and case at high temperature shall be 1,000 megohms minimum.

Dielectric withstanding voltage: 1/

	Sea level V rms (60 Hz)	Altitude V rms (60 Hz)
Between case, frame, or enclosure and all contacts both in the energized and deenergized positions	500	125 All terminals to case
Between case, frame, or enclosure and coil	500	
Between all contacts and coil	500	
Between open contacts in the energized and deenergized positions	500	
Between contact poles	500	

Environmental data:

Temperature range: -65°C to +125°C.

Thermal shock: Applicable.

(D) Shock (specified pulse): MIL-STD-202, method 213, test condition B (75 g's). Contact chatter shall not exceed 10 microseconds maximum for closed contacts and 1 microsecond maximum closure for open contacts.

(D) Vibration (sinusoidal): MIL-STD-202, method 204. Contact chatter shall not exceed 10 microseconds maximum for closed contacts and 1 microsecond maximum closure for open contacts.

1/ WARNING. When testing in the energized position, test voltages shall not be applied across the input leads. In the deenergized position, all leads shall be shorted together

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Magnetic interference: Applicable.

Resistance to soldering heat: Applicable.

Acceleration: Applicable.

Salt atmosphere (corrosion): MIL-STD-750, method 1041.

Physical data:

Terminal strength. MIL-STD-202, method 211.

Pull test: 1-pound pull minimum.

Bend test: 1/2-pound load minimum.

Twist test: As specified in MIL-R-28776.

Solderability: Applicable.

Dimensions and configuration: See figures 1 and 2.

Weight 5.11 grams (0.18 ounce) maximum.

Seal: Hermetic.

Internal moisture: Applicable.

Minimum marking: Military part number, "J" with the date code (example J8530), circuit diagram, manufacturer's name or source code.

Life test requirements:

High level: 100,000 cycles per relay.

Low level: 100,000 cycles plus 900,000 cycles mechanical life.

Part or identifying number (PIN): M28776/7- (dash number from table I and suffix letter designating failure rate level).

Qualification inspection:

Qualification inspection and sample size: See table II.

Initial qualification of relays supplied with mounting pads (-007 through -012) shall be tested as specified below:

Perform the following tests as specified in the qualification inspection table of MIL-R-28776 in the order shown below:

Before installation of pad; screening, visual and mechanical inspection (internal), voltage transients, thermal shock, resistance to solvents, shock (specified pulse), vibration, acceleration, terminal strength, magnetic interference, capacitance (when specified), coil life (applicable to continuous duty relays only), resistance to soldering heat, salt spray (corrosion), overload (applicable to high level relays only), life, terminal strength, and intermediate current.

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TABLE I. Dash numbers and characteristics. 1/

Dash numbers 2/		Supply voltage (V dc) 4/		At +25°C						Over temperature range		
				Coil resistance (ref. only) ohms 5/	Coil current (mA) @ rated 5/ voltage		Pickup voltage (V dc)	Hold voltage (V dc)	Dropout voltage (V dc)	Pickup voltage (V dc)	Hold voltage (V dc)	Drop-out voltage (V dc)
.500 minimum leads	Mounting pads 3/ (fig. 2)	Rated	Max		Max	Min						
001	007	5.0	5.6	100	56.0	43.0	2.9	1.4	0.23	4.0	2.3	0.13
002	008	6.0	8.0	200	33.0	27.0	3.5	2.0	0.32	4.9	3.2	0.18
003	009	9.0	12.0	400	26.4	17.8	5.3	3.0	0.48	7.3	4.9	0.27
004	010	12.0	16.0	800	17.7	11.3	7.1	4.0	0.65	9.8	6.5	0.36
005	011	18.0	24.0	1600	13.8	8.4	10.6	6.0	0.97	14.6	9.8	0.54
006	012	26.5	32.0	3200	10.2	5.8	14.2	8.0	1.30	19.5	13.0	0.72

1/ Each relay possesses high level and low level capabilities. However, relays previously tested or used above 10 mA resistive at 6 V dc maximum or peak ac open circuits not recommended for subsequent use in low level applications.

2/ The suffix letter L, M, P, or R to designate the applicable failure rate level shall be added to the applicable listed dash number. Failure rate level (percent per 10,000 cycles): L, 3.0; M, 1.0; P, 0.1; R, 0.01.
Example, 001L - - - - - - - - - - - 006R.

3/ Relays supplied with mounting pads (-007 through -012) shall have the pad rigidly attached.

4/ CAUTION: The use of any coil supply voltage less than the rated coil supply voltage will compromise the operation of the relay.

5/ Coil resistance not directly measurable at relay terminals. When rated voltage is applied to the coil terminals, the coil current must be within the limits specified. Measure at 25°C at nominal voltage for 5 seconds maximum.

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TABLE II. Qualification inspection and sample size.

Single submission	Group submission	
18 units plus 1 unsealed unit for level L at C = 0 1/ 33 units plus 1 unsealed unit for level M at C = 0 1/ Qualification inspection as applicable.	M28776/7-006	18 units plus 1 unsealed unit for level L at C = 0 1/ 33 units plus 1 unsealed unit for level M at C = 0 1/ Qualification inspection as applicable.
	M28776/7-001 M28776/7-002 M28776/7-003 M28776/7-004 M28776/7-005	2 units each part number Qualification inspection, Group I.

1/ The number of units required for qualification testing will be increased as required in group IV, table II, MIL-R-28776, if the relay manufacturer elects to test the number of units permitting one or more failures. Prior to performance of qualification testing, the relay manufacturer shall preselect the sampling plan.

After installation of pad, perform the following tests as specified in the qualification inspection table of MIL-R-28776, in the order shown below:

Insulation resistance, dielectric withstanding voltage, static contact resistance, pickup voltage, hold voltage, dropout voltage, coil current and coil resistance, turn-on voltage, turn-off voltage, turn-on time and turn-off time, contact dynamic characteristics, and coil transient suppression.

Qualification inspection (reduced testing for previously qualified relays) for relays supplied with mounting pads (-007 through -012). Two units of the 26.5 volt rated coil voltage (-012) shall be tested as specified below:

Before installation of pad, perform the following tests as specified in the qualification inspection table of MIL-R-28776, in the order shown below:

For failure rate level L only: Screening.

For failure rate levels M, P, and R: Vibration (sinusoidal), test duration shall be 10 minutes; particle impact noise detection (P.I.N.D., when specified), and screening.

After installation of pad, perform the following tests as specified in the qualification inspection table of MIL-R-28776, in the order shown below:

Insulation resistance, dielectric withstanding voltage, static contact resistance, pickup voltage, hold voltage, dropout voltage, coil current and coil resistance, turn-on voltage, turn-off voltage, turn-on time and turn-off time, contact dynamic characteristics, and coil transient suppression.

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Group A testing for relays supplied with mounting pads (-007 through -012) shall be tested as specified below:

Before installation of pad, perform subgroup 1 of group A tests.
After installation of pad, perform subgroups 2 and 3 of group A tests.

Qualification inspection (reduced testing based upon similarity): If the relays produced for MIL-R-28776/7 have the same qualified electromechanical internal components as produced for MIL-R-39016/42 or MIL-R-39016/43, then qualification of MIL-R-28776/7 shall be granted following successful completion of the tests listed in table II, groups I, II, and III of the qualification inspection table with the exception of the following tests: Resistance to solvents, acceleration, magnetic interference, and coil life. The number of units to be tested shall be as specified in table III.

TABLE III. Qualification inspection (reduced testing based upon similarity).

Dash number M28776/7-	Number of units
006	8 units plus one open unit for visual inspection for qualification to failure rate level "L" (21 additional units life testing for qualification to failure rate level "M", 0 failures)
001 002 003 004 005	2 units each dash number tested to the screening and electrical characteristics as specified in table II of the qualification inspection table

CONCLUDING MATERIAL

Custodians:
Navy - EC
Air Force - 85

Review activity:
DLA - ES

Preparing activity:
Navy - EC

Agent:
DLA - ES

(Project 5945-0788-02)

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